

Constraint Inventory for the Exact Certification of the Level Profile $\zeta(a) = (5a - 4)/4$

Companion documentation to Appendix A
of “A One-Hypothesis Reduction for Primes in $[4n^2 - n, 4n^2 + n]$ ”

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Abstract

This note is the full constraint inventory referred to in Appendix A of the paper referenced above (henceforth [P]). It derives every linear inequality appearing in the certification script `fm_exact.py` from the published theorem that inequality encodes; states the sign, negation, and boundary conventions needed to read the code’s coefficient tuples; records the three places where the encoding is a deliberate *relaxation* of the true tool, and proves that the certification remains sound there by a sandwich against the Coverage Lemma of [P]; and cross-references every atom family to the code identifiers and to the corresponding statements of [P]. This is designed to be an audit trail, not new mathematics.

1 Purpose and provenance

The script `fm_exact.py` certifies, in exact rational arithmetic, the level profile

$$\zeta(\gamma) = \frac{6 - 5\gamma}{4} \quad (\gamma \in [1, 6/5]),$$

in both directions: for $\beta < \zeta(\gamma)$ every estimation block admits an admissible configuration (achievability — proved by hand in [P], Coverage Lemma, and re-verified here as a byproduct), and at $\beta = \zeta(\gamma)$ a block exists that no encoded tool covers (optimality — [P], Optimality Proposition). The certification is a finite computation because every tool’s admissibility region is a conjunction of inequalities that are *linear* in the block coordinates at fixed γ . This document is the dictionary between those inequalities, the code, and the literature.

2 Conventions

Convention 2.1 (coordinates and exponents). A slice is fixed by $\gamma \in [1, 6/5]$ (prime block $p \sim N^\gamma$; equivalently $a = 2 - \gamma$, cofactor $q \asymp N^a$). The unknowns are $x = (\mu, \eta, \beta)$: the Vaughan-piece exponent, the dyadic frequency exponent, and the sieve-level exponent. All exponents are with respect to base $N = 2n$; $x_{\text{arith}} := 4n^2 \asymp N^2$. We write $\Sigma := \beta + \gamma$ and $\xi := \eta + 2 - \Sigma$ (phase-size exponent). The admissibility requirement for a block, [P] eq. (trilinear requirement), is that the estimating bound have exponent at most Σ (i.e. relative saving N^η

against the trivial size $N^{\eta+\Sigma}$), with an $x^{-c\varepsilon}$ margin suppressed throughout this document: all atoms are stated with $\varepsilon = 0$ and *closed* inequalities. Facet equalities are therefore boundary points of both the pass and fail regions; this is intentional (see Convention 2.3).

Convention 2.2 (atom format in the code). In `failure_atoms()` a constraint is the tuple $\mathcal{C}(a_\mu, a_\eta, a_\beta, c_0, c_\gamma)$, meaning

$$a_\mu \mu + a_\eta \eta + a_\beta \beta \leq c_0 + c_\gamma \gamma,$$

all entries rational. Each *tool* passes on a conjunction of pass conditions; the code stores, per tool, the list of *negated* pass conditions (“fail atoms”), one of which must hold for the tool to fail at a point. A *failure witness* is a point where every tool fails; distributing the disjunctions gives the product over tools of the fail-atom counts:

$$\underbrace{3}_{\text{FI0}} \times \underbrace{3}_{\text{FI1}} \times \underbrace{3}_{\text{FI2}} \times \underbrace{2}_{\text{RS0}} \times \underbrace{2}_{\text{RS1}} \times \underbrace{2}_{\text{RS2}} \times \underbrace{1}_{\text{RS3}} \times \underbrace{1}_{\text{ELEM}} = 216$$

candidate polytopes per piece type (ELEM is present only for type I, where its single fail option is the conjunction of its three negated clauses; type II has 216 combinations without the final factor). Each polytope comprises the chosen fail atoms plus the region constraints of Section 3 (≈ 14 inequalities in 3 unknowns); the minimal β over it is found by exhaustive vertex enumeration ($\binom{14}{3} = 364$ candidate bases, 3×3 systems solved by Cramer’s rule over $\mathbb{Q}$ in `solve3()`), and $\zeta(\gamma)$ is the minimum over all polytopes (`min_beta()`).

Convention 2.3 (boundaries and monotonicity). Because all fail atoms are closed, the computed $\zeta(\gamma)$ is the infimum of levels admitting a witness, and the witness printed at $\beta = \zeta(\gamma)$ sits on the facet with equalities active. Failure is not a priori monotone in β (raising β widens the frequency range and tightens windows, but also widens the reachable grouping intervals), so the script’s output is, by itself, only the statement “the *first* failure occurs at $\beta = \zeta(\gamma)$.” That everything strictly below the facet passes is the Coverage Lemma of [P], proved by hand; the two artifacts meet at the facet. See also Section 7.

3 The block model and the region R

Per [P] (Decomposition and grouping), a block carries variables (h, e_1, e_2, m_v, k) with exponents $(\eta, \beta - s, s, \mu, \gamma - \mu)$, $s \in [0, \beta]$ free. Vaughan pieces: type I has $\mu \in [0, 2\gamma/3]$ with k smooth; type II has $\mu \in [\gamma/3, \gamma/2]$, both coefficients arbitrary. The frequency range from the pipeline lemma is $0 \leq \eta \leq \Sigma - 1$ (blocks with $\Sigma \leq 1$ require nothing). The region constraints in `region()`:

code tuple	inequality	role	piece type
$\mathcal{C}(-1, 0, 0, 0, 0)$	$\mu \geq 0$	piece range	I
$\mathcal{C}(1, 0, 0, 0, 2/3)$	$\mu \leq 2\gamma/3$	piece range	I
$\mathcal{C}(-1, 0, 0, 0, -1/3)$	$\mu \geq \gamma/3$	piece range	II
$\mathcal{C}(1, 0, 0, 0, 1/2)$	$\mu \leq \gamma/2$	piece range	II
$\mathcal{C}(0, -1, 0, 0, 0)$	$\eta \geq 0$	frequency range	both
$\mathcal{C}(0, 1, -1, -1, 1)$	$\eta \leq \beta + \gamma - 1$	frequency range	both
$\mathcal{C}(0, 0, -1, 0, 0), \mathcal{C}(0, 0, 1, 1, 0)$	$0 \leq \beta \leq 1$	search box	both

4 Tool family RS: the Robert–Sargos theorem

4.1 Statement and specialization

Robert–Sargos [3, Theorem 1]: for exponents with $\alpha(\alpha-1)\beta_0\gamma_0 \neq 0$, dyadic ranges H, N_1, M_1 , phase size $X \geq 1$, and arbitrary coefficients $|a(h, \nu)| \leq 1$, $|b(w)| \leq 1$,

$$\sum_{h \sim H} \sum_{\nu \sim N_1} a(h, \nu) \sum_{w \sim M_1} b(w) e\left(X \frac{h^{\beta_0} \nu^{\gamma_0} w^{\alpha}}{H^{\beta_0} N_1^{\gamma_0} M_1^{\alpha}}\right) \ll_{\varepsilon} (HN_1M_1)^{1+\varepsilon} \left\{ \left(\frac{X}{HN_1M_1^2}\right)^{1/4} + (HN_1)^{-1/4} + M_1^{-1/2} + X^{-1/2} \right\}.$$

We use $(\beta_0, \gamma_0, \alpha) = (1, -1, -1)$, so $\alpha(\alpha-1)\beta_0\gamma_0 = -2 \neq 0$; the standalone slot w has exponent m_1 , the joint slot (h, ν) has exponents $(\eta, \Sigma - m_1)$, and $X = N^{\xi}$ with $\xi \geq 2 - \Sigma \geq 3/4 > 0$.

4.2 Window derivation

The requirement is $(\eta + \Sigma) + \max_i t_i \leq \Sigma$, i.e. $\max_i t_i \leq -\eta$, where in exponents, with $n_1 = \Sigma - m_1$ and $\xi - \eta = 2 - \Sigma$:

$$t_1 = \frac{\xi - \eta - n_1 - 2m_1}{4} = \frac{2 - 2\Sigma - m_1}{4} \Rightarrow m_1 \geq 4\eta + 2 - 2\Sigma \quad (\text{A})$$

$$t_2 = -\frac{\eta + n_1}{4} \Rightarrow m_1 \leq \Sigma - 3\eta \quad (\text{B})$$

$$t_3 = -\frac{m_1}{2} \Rightarrow m_1 \geq 2\eta \quad (\text{C})$$

$$t_4 = -\frac{\xi}{2} \Rightarrow \eta \leq 2 - \Sigma \quad (\text{D})$$

Subsumptions used by the encoding: (A) $\Leftarrow$ (C) since $4\eta + 2 - 2\Sigma \leq 2\eta \iff \eta \leq \Sigma - 1$, which is a region constraint; (D) is implied on the relevant part of the region (see Remark 4.1). The effective window is $m_1 \in [2\eta, \Sigma - 3\eta]$, matching [P] (Admissibility window Lemma), whose window is nonempty iff $5\eta \leq \Sigma$.

4.3 The four grouping families and their code atoms

The reachable standalone exponents are $m_1 \in \{s, \mu + s, \gamma - \mu + s, \gamma + s\}$, $s \in [0, \beta]$ ([P], available-set display; the fourth family is retained here because the RS window's upper cap $\Sigma - 3\eta$ can exceed γ at small η). Taking in each family the choice $m_1 = \max(\text{base}, 2\eta)$ where possible, the pass conditions and their negations:

tool	pass conditions	fail atoms (code)	fail inequality
RS0 ($m_1 = s$)	$2\eta \leq \beta; 5\eta \leq \Sigma$	$\mathbf{C}(0, -2, 1, 0, 0)$	$\beta \leq 2\eta$
		$\mathbf{C}(0, -5, 1, 0, -1)$	$\beta \leq 5\eta - \gamma$ (i.e. $5\eta \geq \Sigma$)
RS1 ($m_1 = \mu + s$)	$\mu + \beta \geq 2\eta; \mu \leq \Sigma - 3\eta$	$\mathbf{C}(1, -2, 1, 0, 0)$	$\mu + \beta \leq 2\eta$
		$\mathbf{C}(-1, -3, 1, 0, -1)$	$\mu \geq \Sigma - 3\eta$
RS2 ($m_1 = \gamma - \mu + s$)	$\gamma - \mu + \beta \geq 2\eta; \gamma - \mu \leq \Sigma - 3\eta$	$\mathbf{C}(-1, -2, 1, 0, -1)$	$\gamma - \mu + \beta \leq 2\eta$
		$\mathbf{C}(1, -3, 1, 0, 0)$	$\mu + \beta \leq 3\eta$ (i.e. $\gamma - \mu \geq \Sigma - 3\eta$)
RS3 ($m_1 = \gamma + s$)	$3\eta \leq \beta$ (i.e. $\gamma \leq \Sigma - 3\eta$)	$\mathbf{C}(0, -3, 1, 0, 0)$	$\beta \leq 3\eta$

Remark 4.1 (documented relaxations in the RS encoding). Three omissions make the encoded tools slightly *stronger* than the true ones on part of the search box: (i) **RS1** and **RS2** omit the window-nonemptiness condition $5\eta \leq \Sigma$ (which **RS0** includes); (ii) **RS3** omits the lower-edge

condition $\gamma + s \geq 2\eta$, vacuous only when $2\eta \leq \gamma$, i.e. when $\Sigma \leq (\gamma + 2)/2$; (iii) condition (D) of Subsection 4.2 is omitted everywhere, valid only when $\eta \leq 2 - \Sigma$, i.e. when $\Sigma \leq 3/2$ given $\eta \leq \Sigma - 1$. All three omitted conditions *hold automatically* on $\{\Sigma \leq 5/4, \eta \leq \Sigma - 1\}$, which contains the entire facet region; outside it they are genuine relaxations. Soundness is unaffected: see Section 7.

5 Tool family ELEM: elementary bounds (type I only)

For a type I block the smooth variable $k \sim K = N^\kappa$, $\kappa = \gamma - \mu$, is summed with the phase $f(k) = hz'/(ck)$, $c = e_1 e_2 m_v \sim N^\chi$, $\chi = \beta + \mu$, all other variables in absolute value; $|f'| \asymp \Lambda = N^\lambda$, $\lambda = \eta + 2 - \chi - 2\kappa$. Three clauses, whose disjunction is the pass condition; the tool fails only if all three fail, which is the single ELEM conjunction in the code.

(E1) *Kuzmin–Landau branch* ($\lambda < 0$): by Kuzmin–Landau (e.g. [2, Theorem 2.1], after dyadic subdivision so that $|f'| \leq \frac{1}{2}$), $\sum_k e(f(k)) \ll \Lambda^{-1}$; total exponent $\eta + \chi - \lambda = 2(\chi + \kappa) - 2 = 2\Sigma - 2 \leq \Sigma$ for $\Sigma \leq 2$: *always admissible in region*. Pass clause: $\lambda < 0 \iff \eta \leq \beta - \mu + 2\gamma - 2$.

Fail atom $\mathbf{C}(-1, -1, 1, 2, -2)$: $\eta \geq \beta - \mu + 2\gamma - 2$.

(E2) *Second-derivative test / exponent pair* $(\frac{1}{2}, \frac{1}{2})$ [2, Theorem 2.2]: $\sum_k \ll (\Lambda K)^{1/2} + (K/\Lambda)^{1/2}$. The first term needs $\eta + \chi + \frac{\lambda + \kappa}{2} \leq \Sigma$, which simplifies to $3\eta + 2\mu \leq \beta + 3\gamma - 2$; the companion term needs $\eta + \chi + \frac{\kappa - \lambda}{2} = \frac{\eta + 3\Sigma - 2}{2} \leq \Sigma$, i.e. $\eta \leq 2 - \Sigma$, automatic on the facet region.

Fail atom $\mathbf{C}(-2, -3, 1, 2, -3)$: $3\eta + 2\mu \geq \beta + 3\gamma - 2$.

(E3) *Exponent pair* $(\frac{1}{6}, \frac{2}{3})$ [2, §3]: $\sum_k \ll \Lambda^{1/6} K^{2/3} (+ \text{admissible companions as in (E1), (E2)})$; the main term needs $\eta + \chi + \frac{\lambda}{6} + \frac{2\kappa}{3} = \frac{7\eta + 5\beta + 3\mu + 2\gamma + 2}{6} \leq \Sigma$, which simplifies to $7\eta + 3\mu \leq \beta + 4\gamma - 2$.

Fail atom $\mathbf{C}(-3, -7, 1, 2, -4)$: $7\eta + 3\mu \geq \beta + 4\gamma - 2$.

Remark 5.1 (P). uses only (E1) and (E2) in the Coverage Lemma; (E3) is carried in the certification solely so that the Optimality Proposition quantifies over the larger toolkit. The certified optimum is identical with or without (E3) — one of the two byproducts recorded in [P], Appendix A.

6 Tool family FI: Fouvry–Iwaniec Theorem 6

This family is *not used* in the proofs of [P]; it is encoded so that the Optimality Proposition's claim — the profile is unchanged when [1, Theorem 6] is adjoined — is machine-checked rather than asserted. Since the derivation appears nowhere in [P], it is recorded here in full.

6.1 Statement and exponent form

[1, Theorem 6]: for $\alpha \neq 0, 1$, an additive character $\chi(h)$, and arbitrary $|\varphi_m|, |\psi_n| \leq 1$,

$$S_{\chi\varphi\psi}(H, M, N) \ll (HMN)^{1/2} \left[(H + N)^{1/2} (x_6^{1/8} H^{-1/6} M^{1/12} N^{1/6} + x_6^{1/8} H^{-1/8} N^{3/8} + N^{1/2} + N^{1/4} M^{1/8}) x_6^{1/8} + M^{1/2} + x_6^{-1/4} M^{1/2} N \right] \mathcal{L}^4,$$

where x_6 is the phase size. With slot exponents (m_0, n_0) , $m_0 + n_0 = \Sigma$, frequency η , and $x_6 = N^\xi$, the requirement (bound $\leq N^\Sigma$) yields, after the prefactor $(\eta + \Sigma)/2$:

- the $M^{1/2}$ term (T5): $\frac{\eta+\Sigma}{2} + \frac{m_0}{2} \leq \Sigma \iff \eta \leq n_0$ — a *lower* window edge, which also fixes $(H + N)^{1/2} \mapsto N^{n_0/2}$ in the remaining terms;
- five upper caps on n_0 , derived by direct substitution ($\xi = \eta + 2 - \Sigma$, $m_0 = \Sigma - n_0$):

$$\begin{aligned}
\text{T1 } (x_6^{1/4} H^{-1/6} M^{1/12} N^{1/6}.): \quad & 7n_0 \leq 8\Sigma - 7\eta - 6 \Rightarrow n_0 \leq \text{cap}_1 = \frac{8\Sigma - 7\eta - 6}{7} \\
\text{T2 } (x_6^{1/4} H^{-1/8} N^{3/8}.): \quad & 7n_0 \leq 6\Sigma - 5\eta - 4 \Rightarrow n_0 \leq \text{cap}_2 = \frac{6\Sigma - 5\eta - 4}{7} \\
\text{T3 } (x_6^{1/8} N^{1/2}.): \quad & 8n_0 \leq 5\Sigma - 5\eta - 2 \Rightarrow n_0 \leq \text{cap}_3 = \frac{5\Sigma - 5\eta - 2}{8} \\
\text{T4 } (x_6^{1/8} N^{1/4} M^{1/8}.): \quad & 5n_0 \leq 4\Sigma - 5\eta - 2 \Rightarrow n_0 \leq \text{cap}_4 = \frac{4\Sigma - 5\eta - 2}{5} \\
\text{T6 } (x_6^{-1/4} M^{1/2} N.): \quad & 2n_0 \leq 2 - \Sigma - \eta \Rightarrow n_0 \leq \text{cap}_6 = \frac{2 - \Sigma - \eta}{2}
\end{aligned}$$

(Sample derivation, T2: multiply the requirement $\frac{\eta+\Sigma}{2} + \frac{n_0}{2} + \frac{\xi}{4} - \frac{\eta}{8} + \frac{3n_0}{8} \leq \Sigma$ by 8 and substitute ξ : $5\eta + 7n_0 + 2\Sigma + 4 \leq 8\Sigma$. The others are identical in kind.)

6.2 Pruning lemma

Lemma 6.1. *On $\{\eta \leq \Sigma - 1, \Sigma \leq 4/3\}$: $\text{cap}_2 \leq \text{cap}_1$, $\text{cap}_2 \leq \text{cap}_3$, and $\text{cap}_2 \leq \text{cap}_4$. Hence the FI window is $[\eta, \min(\text{cap}_6, \text{cap}_2)]$.*

Proof. $\text{cap}_1 - \text{cap}_2 = \frac{2\Sigma - 2\eta - 2}{7} \geq 0$ iff $\eta \leq \Sigma - 1$. $\text{cap}_2 \leq \text{cap}_4$ iff $5(6\Sigma - 5\eta - 4) \leq 7(4\Sigma - 5\eta - 2)$ iff $\Sigma + 5\eta \leq 3$; with $\eta \leq \Sigma - 1$ this follows from $6\Sigma \leq 8$, i.e. $\Sigma \leq 4/3$. $\text{cap}_2 \leq \text{cap}_3$ iff $8(6\Sigma - 5\eta - 4) \leq 7(5\Sigma - 5\eta - 2)$ iff $13\Sigma \leq 5\eta + 18$, implied by $\Sigma \leq 4/3 < 18/13$. $\square$

6.3 Grouping families and code atoms

The reachable n_0 families are $\{s, \mu + s, \gamma - \mu + s\}$, $s \in [0, \beta]$; the fourth family $n_0 = \gamma + s$ is *provably dead* for this tool, since $\gamma \geq 1$ while $\text{cap}_6 \leq \frac{2-\Sigma}{2} \leq \frac{1}{2}$, and is therefore not encoded. In each family we take n_0 at the base value (lower edge condition base $+\beta \geq \eta$):

tool	pass conditions	fail atoms (code)	fail inequality
FI0	$\eta \leq \beta; \eta \leq \text{cap}_6; \eta \leq \text{cap}_2$	$C(0, -1, 1, 0, 0)$	$\eta \geq \beta$
		$C(0, -3, -1, -2, 1)$	$3\eta + \beta \geq 2 - \gamma \quad (\eta \geq \text{cap}_6)$
		$C(0, -12, 6, 4, -6)$	$12\eta - 6\beta \geq 6\gamma - 4 \quad (\eta \geq \text{cap}_2)$
FI1	$\mu + \beta \geq \eta; \mu \leq \text{cap}_6; \mu \leq \text{cap}_2$	$C(1, -1, 1, 0, 0)$	$\mu + \beta \leq \eta$
		$C(-2, -1, -1, -2, 1)$	$2\mu + \eta + \beta \geq 2 - \gamma$
		$C(-7, -5, 6, 4, -6)$	$7\mu + 5\eta - 6\beta \geq 6\gamma - 4$
FI2	$\gamma - \mu + \beta \geq \eta; \gamma - \mu \leq \text{cap}_6; \gamma - \mu \leq \text{cap}_2$	$C(-1, -1, 1, 0, -1)$	$\gamma - \mu + \beta \leq \eta$
		$C(2, -1, -1, -2, 3)$	$2\mu - \eta - \beta \leq 3\gamma - 2$
		$C(7, -5, 6, 4, 1)$	$7\mu - 5\eta + 6\beta \leq \gamma + 4$

(The last two FI2 rows are the sign-flipped forms of $\gamma - \mu \geq \text{cap}_6$ and $\gamma - \mu \geq \text{cap}_2$.) The pruning is a further relaxation outside $\Sigma \leq 4/3$, handled by the same soundness argument below. The trivial-size lower edge $\eta \leq n_0$ is encoded as the first pass condition in each family.

7 Soundness of the relaxed encoding

Let $\zeta_{\text{code}}(\gamma)$ be the script's output and $\zeta_{\text{true}}(\gamma)$ the first-failure level of the *exactly* encoded toolkit. Every deviation documented in Remark 4.1 and Subsection 6.2 makes an encoded tool pass on a superset of its true pass region; failure witnesses are therefore harder to find, so $\zeta_{\text{code}} \geq \zeta_{\text{true}}$. In the other direction, the Coverage Lemma of [P] proves by hand — using only the exactly-valid parts of the encoding, all active inside $\{\Sigma \leq 5/4\}$ where every omitted condition holds — that all blocks pass for $\beta < (6 - 5\gamma)/4$, so $\zeta_{\text{true}} \geq (6 - 5\gamma)/4$. Since the script returns $\zeta_{\text{code}} = (6 - 5\gamma)/4$ exactly at every tested rational γ , the sandwich closes:

$$\frac{6 - 5\gamma}{4} = \zeta_{\text{code}} \geq \zeta_{\text{true}} \geq \frac{6 - 5\gamma}{4}.$$

Finally, the printed witnesses are legitimate witnesses for the *true* system: at $(\mu, \eta) = (\gamma/2, \Sigma - 1)$ with $4\beta + 5\gamma = 6$ one checks directly that every omitted condition ($5\eta \leq \Sigma$, $\eta \leq 2 - \Sigma$, $2\eta \leq \gamma$, $\Sigma \leq 4/3$) is satisfied, so no relaxed tool was wrongly credited there.

8 Cross-reference table

code identifier	this document	item in [P]
RS0–RS3	§4	§5.2 Admissibility-window Lemma; available-set display
ELEM clause (E1)	§5	§5.5 elementary Lemma (Kuzmin–Landau branch)
ELEM clause (E2)	§5	§5.6 elementary Lemma (main facet)
ELEM clause (E3)	§5	§5.5 certification only (Optimality Prop.)
F10–F12	§6	§5.5 certification only (Optimality Prop.; Appendix A byproducts)
region()	§3	§5.3 Decomposition-and-grouping subsection; pipeline Lemma
solve3(), min_beta()	Convention 2.2	§A Appendix A (vertex enumeration)

9 Reproduction

Run `python3 fm_exact.py` (Python 3; `numpy/scipy` are used only for the final floating-point root-finding of θ^{**} , which is verification of a closed-form calculus step and not part of the certification). Expected output: the exact table $\zeta(\gamma) = 1/4, 3/16, 1/8, 1/16, 0$ at $\gamma = 1, 21/20, 11/10, 23/20, 6/5$, agreement with $(6 - 5\gamma)/4$ throughout, the witness $(\mu, \eta) = (11/20, 9/40)$ at $\gamma = 11/10$, the threshold $\theta^{**} = 0.939305$, and the margins $+0.0136, +0.2028, +0.5478$ (in units of u) at $\vartheta = 0.94, 0.95, 0.97$. Runtime is a few minutes.

References

- [1] E. Fouvry and H. Iwaniec, *Exponential sums with monomials*, J. Number Theory **33** (1989), 311–333.
- [2] S. W. Graham and G. Kolesnik, *Van der Corput's Method of Exponential Sums*, London Math. Soc. Lecture Note Ser. **126**, Cambridge Univ. Press, 1991.
- [3] O. Robert and P. Sargos, *Three-dimensional exponential sums with monomials*, J. reine angew. Math. **591** (2006), 1–20.